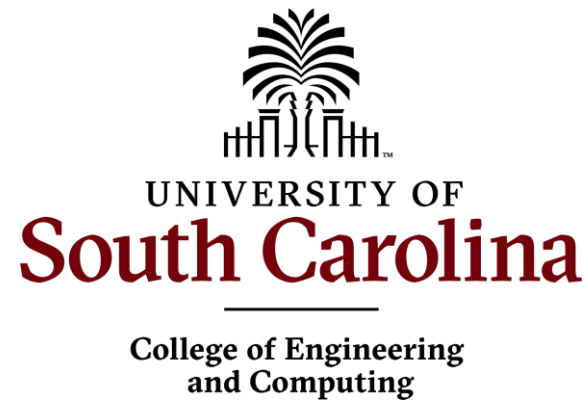


INTRODUCTION TO APPLIED NUMERICAL METHODS

Gauss Elimination

Lang Yuan, Associate Professor



MATRIX OPERATING RULES (MULTIPLICATION)

Product of Two Matrices

$$[C] = [A] \cdot [B] \quad \Rightarrow \quad c_{ij} = \sum_{k=1}^m a_{ik} b_{kj}$$

m = the column dimension of $[A]$ and the row dimension of $[B]$

$$[A]_{n \times m} \quad [B]_{m \times l} = [C]_{n \times l}$$

Interior dimensions are equal; multiplication is possible

Exterior dimensions define the dimensions of the result

the number of columns of the first matrix must be equal to the number of rows (m) in the second matrix

REPRESENTING LINEAR ALGEBRAIC EQUATIONS IN MATRIX FORM

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n &= b_1 \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n &= b_2 \\ &\vdots \\ a_{n1}x_1 + a_{n2}x_2 + \cdots + a_{nn}x_n &= b_n \end{aligned}$$

Linear algebraic equations

$$[A][X] = [B]$$

Matrix Form

$$[A] = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{bmatrix}$$

[A] is the n by n square matrix of coefficients

$$[X] = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_{n-1} \\ x_n \end{bmatrix}$$

[X] is the n by 1 column vector of unknowns

$$[B] = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_{n-1} \\ b_n \end{bmatrix}$$

[B] is the n by 1 column vector of constants

REPRESENTING LINEAR ALGEBRAIC EQUATIONS IN MATRIX FORM

Prove

$$x_1 + 2x_2 + 3x_3 = 4$$

$$3x_1 + 2x_2 + x_3 = 5$$

$$4x_1 + x_2 + 3x_3 = 6$$



$$\begin{bmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \\ 4 & 1 & 3 \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \\ x_3 \end{Bmatrix} = \begin{Bmatrix} 4 \\ 5 \\ 6 \end{Bmatrix}$$

[A] [X] [B]

Step 1: Matrix multiplication using the matrix form

$$[A] = \begin{bmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \\ 4 & 1 & 3 \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \\ x_3 \end{Bmatrix} = \begin{Bmatrix} b_1 \\ b_2 \\ b_3 \end{Bmatrix}$$

[X] [B]

$$\begin{aligned} b_1 &= x_1 + 2x_2 + 3x_3 \\ b_2 &= 3x_1 + 2x_2 + x_3 \\ b_3 &= 4x_1 + x_2 + 3x_3 \end{aligned}$$

$$[B] = \begin{Bmatrix} 4 \\ 5 \\ 6 \end{Bmatrix}$$

$$\begin{aligned} x_1 + 2x_2 + 3x_3 &= 4 \\ 3x_1 + 2x_2 + x_3 &= 5 \\ 4x_1 + x_2 + 3x_3 &= 6 \end{aligned}$$

Step 2: Because

SPECIAL MATRIX: SQUARE MATRIX

Matrices where $n = m$ are called *square matrices*

$$[A] = \begin{bmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \\ 4 & 1 & 3 \end{bmatrix}$$

$$x_1 + 2x_2 + 3x_3 = 4$$

$$3x_1 + 2x_2 + x_3 = 5$$

$$4x_1 + x_2 + 3x_3 = 6$$



$$\begin{bmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \\ 4 & 1 & 3 \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \\ x_3 \end{Bmatrix} = \begin{Bmatrix} 4 \\ 5 \\ 6 \end{Bmatrix}$$

- Number of unknowns and the number of equations are equal for unique solution
- Important for solving the set of linear equations

MATRIX TRANSPOSE

- Matrix Transpose
 - Transforming its rows into its columns and its columns into rows
 - Swap a_{ij} and a_{ji}

$$[A] = \begin{bmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ a_{21} & a_{22} & a_{23} & a_{24} \\ a_{31} & a_{32} & a_{33} & a_{34} \\ a_{41} & a_{42} & a_{43} & a_{44} \end{bmatrix} \xrightarrow{\text{Transpose: T}} [A]^T = \begin{bmatrix} a_{11} & a_{21} & a_{31} & a_{41} \\ a_{12} & a_{22} & a_{32} & a_{42} \\ a_{13} & a_{23} & a_{33} & a_{43} \\ a_{14} & a_{24} & a_{34} & a_{44} \end{bmatrix}$$

$$[A] = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$$

$$[A]^T = \begin{bmatrix} 1 & 4 & 7 \\ 2 & 5 & 8 \\ 3 & 6 & 9 \end{bmatrix}$$

$$[B] = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_{n-1} \\ b_n \end{bmatrix} \xrightarrow{\quad} [B]^T = [b_1 \quad b_2 \quad \dots \quad b_{n-1} \quad b_n]$$

$$[B] = \begin{bmatrix} 3 \\ 7 \\ 11 \\ 15 \end{bmatrix} \xrightarrow{\quad} [B]^T = [3 \quad 7 \quad 11 \quad 15]$$

MATRIX AUGMENTATION

- Matrix Augmentation
 - Attaching a column (or columns) to the original matrix

$$[A] = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \xrightarrow{\text{Augmentation}} \left[\begin{array}{ccc|ccc} a_{11} & a_{12} & a_{13} & 1 & 0 & 0 \\ a_{21} & a_{22} & a_{23} & 0 & 1 & 0 \\ a_{31} & a_{32} & a_{33} & 0 & 0 & 1 \end{array} \right]$$

Augment [A] with [B]:

$$[A] = \left[\begin{array}{ccc|c} a_{11} & a_{12} & a_{13} & b_1 \\ a_{21} & a_{22} & a_{23} & b_2 \\ a_{31} & a_{32} & a_{33} & b_3 \end{array} \right]$$

- Very useful when a set of identical operations needed to be performed on two matrices (or vectors)

ELIMINATION OF UNKNOWNNS

Approach

- Multiply the equations by constants so that one of the unknowns will be eliminated when the two equations are combined
- A single equation forms and can be solved for the remaining unknown
- Substitute the obtained values back to the equation to solve for the other unknowns

$$\begin{array}{l} a_{11}x_1 + a_{12}x_2 = b_1 \quad \textcircled{1} \\ a_{21}x_1 + a_{22}x_2 = b_2 \quad \textcircled{2} \end{array}$$



Step 1: $\textcircled{1} \times \left(\frac{a_{21}}{a_{11}}\right)$

$$a_{21}x_1 + \frac{a_{12}a_{21}}{a_{11}}x_2 = \frac{b_1a_{21}}{a_{11}} \quad \textcircled{1'}$$

Step 2: $\textcircled{2} - \textcircled{1'}$ $\textcircled{2'}$

$$\left(a_{22} - \frac{a_{12}a_{21}}{a_{11}}\right)x_2 = b_2 - \frac{b_1a_{21}}{a_{11}}$$

ELIMINATION OF UNKNOWNNS

Approach

- Multiply the equations by constants so that one of the unknowns will be eliminated when the two equations are combined
- A single equation results and can be solved for the remaining unknown
- Substitute the obtained values back to the equation to solve for the other unknowns

Step 3:

$$x_2 = \frac{a_{11}b_2 - a_{21}b_1}{a_{11}a_{22} - a_{12}a_{21}}$$

2'

Step 4:

2'

→

1

$$x_1 = \frac{a_{22}b_1 - a_{12}b_2}{a_{11}a_{22} - a_{12}a_{21}}$$

AN EXAMPLE OF ELIMINATION OF UNKNOWNNS

$$3x_1 + 2x_2 = 18$$

$$-x_1 + 2x_2 = 2$$

Solution:

$$\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} = \begin{bmatrix} 3 & 2 \\ -1 & 2 \end{bmatrix} \quad \begin{bmatrix} b_1 \\ b_2 \end{bmatrix} = \begin{bmatrix} 18 \\ 2 \end{bmatrix}$$

$$x_1 = \frac{a_{22}b_1 - a_{12}b_2}{a_{11}a_{22} - a_{12}a_{21}} = \frac{2(18) - 2(2)}{3(2) - 2(-1)} = 4$$

$$x_2 = \frac{a_{11}b_2 - a_{21}b_1}{a_{11}a_{22} - a_{12}a_{21}} = \frac{3(2) - (-1)18}{3(2) - 2(-1)} = 3$$

How to extend the elimination to a large system consisting of more than 2 equations ?

GAUSS ELIMINATION: AN EXAMPLE

Use Gauss elimination to solve

$$3x_1 - 0.1x_2 - 0.2x_3 = 7.85 \quad (1)$$

$$0.1x_1 + 7x_2 - 0.3x_3 = -19.3 \quad (2)$$

$$0.3x_1 - 0.2x_2 + 10x_3 = 71.4 \quad (3)$$

Solution: Forward Elimination

Step 1.1:

Eq. (1) $\times (0.1/3)$ $\textcircled{1} \times \left(\frac{0.1}{3}\right)$

$$0.1x_1 - \frac{0.01}{3}x_2 - \frac{0.02}{3}x_3 = 0.2617 \quad (1^*)$$

Subtract Eq. (1*) from Eq. (2)

$$0.1x_1 + 7x_2 - 0.3x_3 = -19.3 \quad (2)$$

$$0.1x_1 - \frac{0.01}{3}x_2 - \frac{0.02}{3}x_3 = 0.2617 \quad (1^*)$$

11

$$7.00333x_2 - 0.293333x_3 = -19.5617 \quad (2')$$

New Equation Set

$$3x_1 - 0.1x_2 - 0.2x_3 = 7.85 \quad (1)$$

$$7.00333x_2 - 0.293333x_3 = -19.5617 \quad (2')$$

$$0.3x_1 - 0.2x_2 + 10x_3 = 71.4 \quad (3)$$

GAUSS ELIMINATION: AN EXAMPLE

Solution: **Forward Elimination**

Step 1.2:

$$\text{Eq. (1)} \times (0.3/3) \quad \textcircled{1} \times \left(\frac{0.3}{3}\right)$$

$$0.3x_1 - 0.01x_2 - 0.02x_3 = 0.785 \quad (1^+)$$

Subtract Eq. (1⁺) from Eq. (3)

$$0.3x_1 - 0.2x_2 + 10x_3 = 71.4 \quad (3)$$

—

$$0.3x_1 - 0.01x_2 - 0.02x_3 = 0.785 \quad (1^+)$$



$$-0.190000x_2 + 10.0200x_3 = 70.6150 \quad (3')$$



New Equation Set

$$3x_1 - 0.1x_2 - 0.2x_3 = 7.85 \quad (1)$$

$$7.00333x_2 - 0.293333x_3 = -19.5617 \quad (2')$$

$$0.3x_1 - 0.2x_2 + 10x_3 = 71.4 \quad (3)$$

New Equation Set

$$3x_1 - 0.1x_2 - 0.2x_3 = 7.85 \quad (1)$$

$$7.00333x_2 - 0.293333x_3 = -19.5617 \quad (2')$$

$$-0.190000x_2 + 10.0200x_3 = 70.6150 \quad (3')$$

GAUSS ELIMINATION: AN EXAMPLE

New Equation Set

$$\begin{array}{rcl} 3x_1 & -0.1x_2 & -0.2x_3 = 7.85 & (1) \\ & 7.00333x_2 & -0.293333x_3 = -19.5617 & (2') \\ & -0.190000x_2 & + 10.0200x_3 = 70.6150 & (3') \end{array}$$

Solution: Forward Elimination

Step 2:

Eq. (2') $\times (-0.19/7.00333)$ $\left(\begin{smallmatrix} 2 \\ , \end{smallmatrix} \right) \times \left(\frac{-0.19}{7.00333} \right)$

$$-0.19x_2 + 0.0079581x_3 = 0.5307 \quad (2^*)$$

Subtract Eq. (2*) from Eq. (3')

$$-0.190000x_2 + 10.0200x_3 = 70.6150 \quad (3')$$

—

$$-0.19x_2 + 0.0079581x_3 = 0.5307 \quad (2^*)$$



$$10.0120x_3 = 70.0843 \quad (3'')$$



New Equation Set

$$\begin{array}{rcl} 3x_1 & -0.1x_2 & -0.2x_3 = 7.85 \\ & 7.00333x_2 & -0.293333x_3 = -19.5617 \\ & & 10.0120x_3 = 70.0843 \end{array}$$

GAUSS ELIMINATION: AN EXAMPLE

New Equation Set

$$3x_1 - 0.1x_2 - 0.2x_3 = 7.85 \quad (1)$$

$$7.00333x_2 - 0.293333x_3 = -19.5617 \quad (2')$$

$$10.0120x_3 = 70.0843 \quad (3'')$$

Solution: **Back Substitution**

Eq. (3'') can be solved for

$$x_3 = \frac{70.0843}{10.0120} = 7.0000$$

Back-substitute x_3 to Eq. (2')

$$7.00333x_2 - 0.293333(7.0000) = -19.5617$$

$$x_2 = \frac{-19.5617 + 0.293333(7.0000)}{7.00333} = -2.50000$$

GAUSS ELIMINATION: AN EXAMPLE

$$\begin{array}{rcl} 3x_1 & -0.1x_2 & -0.2x_3 = 7.85 \quad (1) \\ & 7.00333x_2 - 0.293333x_3 = -19.5617 & (2') \\ & 10.0120x_3 = 70.0843 & (3'') \end{array}$$

Solution: **Back Substitution**

Back-substitute x_2 and x_3 to Eq. (1)

$$\begin{array}{c} 3x_1 - 0.1x_2 - 0.2x_3 = 7.85 \\ \downarrow \\ 3x_1 - 0.1(-2.50000) - 0.2(7.0000) = 7.85 \\ \downarrow \\ x_1 = \frac{7.85 + 0.1(-2.50000) + 0.2(7.0000)}{3} = 3.00000 \end{array}$$

GAUSS ELIMINATION: AN EXAMPLE

Use Gauss elimination to solve

$$3x_1 - 0.1x_2 - 0.2x_3 = 7.85 \quad (1)$$

$$0.1x_1 + 7x_2 - 0.3x_3 = -19.3 \quad (2)$$

$$0.3x_1 - 0.2x_2 + 10x_3 = 71.4 \quad (3)$$

Verification:

$$x_1 = 3, x_2 = -2.5, \text{ and } x_3 = 7$$



$$3(3) - 0.1(-2.5) - 0.2(7) = 7.85$$

$$0.1(3) + 7(-2.5) - 0.3(7) = -19.3$$

$$0.3(3) - 0.2(-2.5) + 10(7) = 71.4$$

Basic Steps:

- Elimination of unknowns
- Solution through back substitution